

# Design Lab Report

ROB 467 - Group 1

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## I. ABSTRACT

We present Scribbles and Specs, an autonomous robotics system that can erase ink on a whiteboard in addition to drawing images. The system has one robot for perception and one for actuation. Specs, a stationary Raspberry Pi and PiCam module, performs global localization of the board and the rover as well as ink detection. Scribbles, a lightweight mobile rover built on a custom 3D-printed chassis with an ESP32 microcontroller and three DC motors driving omnidirectional wheels, drives between waypoints received over a wireless link. The original design called for a vertically adhering rover using magnets, but with the resources and time available, we were unable to improve our design to produce sufficient normal force to counteract sliding from the omnidirectional wheels. Therefore, we pivoted to a horizontal whiteboard configuration, which preserves all sensing, planning, and communication subsystems while eliminating the adhesion constraint. The resulting prototype demonstrates ink detection, rover localization, wireless command streaming, autonomous erasing, and the autonomous drawing of a provided image. Localization with the april tags is accurate to 0.005 m, and path following is accurate to 0.01m. In tests of erasing, up to 80% of ink on the board was erased. This paper describes the system architecture and design changes made during development, as well as evaluation of the prototype.

**Keywords:** Autonomous robotics, external localization, computer vision, wireless communication, path planning, embedded systems.

## II. INTRODUCTION

Whiteboards are commonly found in collaborative environments, from classrooms and research labs to corporate meeting rooms. Despite their usefulness as a tool for ideation and instruction, all interaction with whiteboards remains manual. Users need to write content by hand, limiting repeatability and precision. Users must also erase content manually, which is oftentimes forgotten.

While autonomous operation has been extensively explored for horizontal surfaces (such as robotic vacuum cleaners), there is no consumer-grade solution for automated drawing on whiteboards. Navigation on a flat, elevated surface introduces a new set of constraints, including workspace boundaries and the need for precise localization relative to a bounded region.

This project proposes a two-robot autonomous robotic system that addresses whiteboard maintenance. By separating the computationally heavy sensing tasks into a stationary

external camera (Specs) and a minimal, lightweight mobile rover (Scribbles), we arrive at a scalable, low-cost architecture in which the perception module can be reused across multiple rovers and board sizes. The primary goals of the system are to detect ink on a whiteboard, localize the rover on the board using an external camera, and wirelessly command the rover to follow a trajectory.

Although the original design targeted a vertical whiteboard surface with magnetic adhesion, testing revealed that our omni-wheel drivetrain did not retain enough friction under magnetic loading to prevent sliding down the whiteboard surface. We therefore pivoted to a horizontal whiteboard, which preserves all perception, planning, and actuation subsystems while sidestepping the adhesion problem for now.

## III. RELATED WORK

Several prior works validate the feasibility of wall-climbing platforms that rely on magnetic adhesion. The MagnetMobile project [1] demonstrates a four-wheeled rover that uses permanent magnets for vertical mobility on ferrous surfaces. Wang et al. [2] explore adaptive magnetic adhesion systems that allow robots to traverse surfaces with varying inclines, including internal corners. Hu et al. [3] present a tracked wall-climbing robot embedded with magnets, showing that track-based contact improves traction on curved surfaces compared to individual wheels.

These works establish that magnetic adhesion is a common and reliable solution for vertical traversal on ferrous surfaces, but they focus almost exclusively on mobility in harsh environments — hulls, pipes, storage tanks — rather than on interaction with a structured human-facing surface. Our system differs in two ways. First, the primary contribution of our platform is not adhesion itself but task-level interaction with a writing surface. Second, we rely on external vision-based localization from a fixed camera rather than onboard inertial or odometric estimation, which simplifies the rover and reduces onboard computational load. Additionally, the adhesion mechanism from these prior works did not transfer well to our wheel geometry and writing-surface constraints, which motivated the move to a horizontal configuration for the present prototype. The thicker wheels and rugged surface offer more wheel-to-surface friction which helped robots stay vertical, whereas in our case the friction between the omnidirectional wheels and slippery whiteboard was significantly lower.

## IV. METHODOLOGY

### A. System Overview

The system consists of two agents that communicate over a Wi-Fi link. Specs, the stationary perception module, is built around a Raspberry Pi and a PiCam mounted so the entire whiteboard is in frame. It is responsible detecting the board and tracking the rover. Scribbles, the mobile agent, is a custom 3D-printed rover driven by an ESP32 microcontroller that executes velocity commands received from Specs. The two agents form a closed sense-plan-act loop in which Specs performs sensing and high-level planning and Scribbles performs low-level planning actuation, so the on-board computational requirements for the rover are minimal.

### B. Scribbles: Mobile Rover

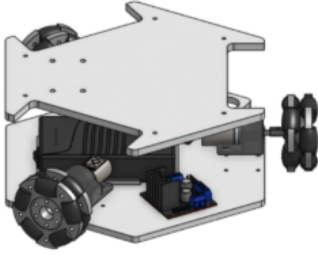


Fig. 1: CAD model of the Scribbles rover

Scribbles is built on a custom 3D-printed chassis. Three DC motors drive omni-directional wheels arranged at 120° intervals around the chassis, giving holonomic motion in the plane of the whiteboard. Two dual H-bridge motor drivers provide bi-directional PWM control of the three motors from an onboard ESP32 microcontroller. An eraser or whiteboard marker may be mounted to the chassis to allow for erasing and drawing, respectively.

The ESP32 runs a compact firmware loop that receives target waypoints from Specs over Wi-Fi, converts them into velocity commands and individual wheel speeds using the standard omni-wheel inverse kinematics, and commands the H-bridge drivers accordingly. Because all localization and planning happen off-board, the rover carries no IMU, encoders, or onboard camera; this keeps Scribbles small, light, and inexpensive.

### C. Specs: External Vision and Planning

Specs is shown in Fig. 2. A Raspberry Pi hosts the perception stack and a PiCam is positioned so that the full whiteboard is in view. The perception stack performs two distinct tasks:

1) Board detection: An April tag is placed at each of the four corners of the whiteboard. Specs detects these markers, computes the homography between the image plane and the whiteboard plane, and uses it to rectify subsequent frames. This gives a known mapping from image pixels to physical coordinates on the board, which is used by all downstream stages.

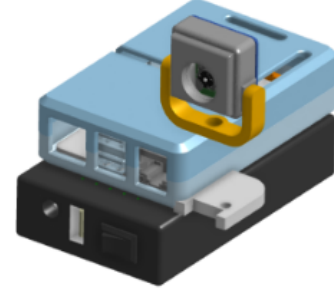


Fig. 2: CAD model of the Specs perception module

2) Rover localization: A fifth April tag is mounted on top of Scribbles. Its pose in the rectified image gives the rover's position and heading on the board. This pose is used as the feedback signal in a simple proportional controller that drives the rover toward the next waypoint.

During drawing, specs is also capable of taking an input image, converting it to a series of waypoints, and sending those waypoints to scribbles to execute.

### D. Planning and Control

Given a target drawing or erasure pattern and the rover's pose, Specs plans a trajectory that traces the desired pattern on the whiteboard. The current implementation converts a user-specified image or path into a sequence of waypoints in board coordinates, for example by rasterizing the image into line segments. Specs sends these waypoints one-by-one to Scribbles, with the next waypoint being sent once Scribbles reaches the current waypoint. Specs also sends Scribbles its pose at roughly 20 Hz.

On Scribbles, path following is performed with the pure pursuit algorithm [4]. This algorithm computes the vector from Scribbles' current position (from the localization performed by Specs) to a lookahead point on the straight-line segment between waypoints Scribbles is following and calculates a velocity  $v_d^w$  along that vector which has speed  $s$ . This world-frame velocity is transformed to the frame of the body of Scribbles:  $v_d^b = R_w^b v_d^w$ . To avoid overshooting waypoints, the speed of scribbles is reduced when close to a waypoint:

$$s = \begin{cases} \frac{|X - X_d|}{0.05} s_0 & |X - X_d| < 0.05 \\ s_0 & |X - X_d| \geq 0.05 \end{cases} \quad (1)$$

Where  $s_0$  is the standard speed of scribbles.

Along the path, Scribbles should maintain orientation, which we define as a heading of 0°. We use P control for this and command an angular velocity  $\omega$  for Scribbles to follow:  $\omega = -K_\omega \theta$  where  $\theta$  is the current heading of scribbles.

The desired spatial and angular velocities of scribbles are translated to individual wheel speeds through inverse kinematics [5]:

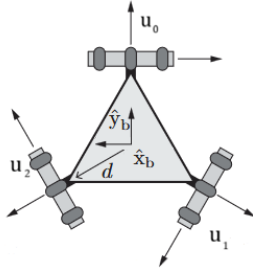


Fig. 3: Wheel numbering and parameters for Scribbles. Adapted from the diagrams in [5].

$$U = \begin{bmatrix} u_0 \\ u_1 \\ u_2 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} d & 1 & 0 \\ d & -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ d & -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} \omega \\ v_x^b \\ v_y^b \end{bmatrix} \quad (2)$$

The desired wheel speed is tracked using a feedforward PWM command and PID command to reduce error. During development, constant PWM inputs were commanded to the wheels and the steady-state wheel speed was recorded. From this a least-squares linear approximation relating PWM command to wheel speed was calculated. This linear approximation is used to calculate the feedforward PWM term. Then, the error-correcting PID term is calculated using wheel speed feedback as measured from encoders. With this controller configuration, tuning becomes much simpler: the PID parameters have a lower effect, and for the pure pursuit controller, the only influential parameter, the lookahead distance, corresponds to how aggressively the robot should return to the straight line path, making tuning directly comparable to visual accuracy for drawing.

#### E. Communication

Specs and Scribbles communicate over Wi-Fi. The ESP32 on Scribbles creates a server that Specs connects to. The Raspberry Pi publishes waypoint commands at 0.5 Hz and localization estimates of the pose at 20 Hz to the ESP32. If no position information is received in 2.5 seconds, Scribbles halts as a safety fallback.

#### F. Materials

The prototype of Scribbles is built from the following components: a custom laser-cut chassis; three DC motors with omni-directional wheels; two dual H-bridge motor driver boards; an ESP32 microcontroller; a whiteboard eraser; a dry-erase marker; and a USB power bank for the rover. Specs is built with a Raspberry Pi with PiCam. On the board there are four April tags one April tag on the rover. Total out-of-pocket cost was under fifty US dollars, with several components reused from the lab and team inventory.

### V. RESULTS

#### A. Evaluation Methodology

We evaluated the integrated system along three axes:

| Axis          | Mean error |
|---------------|------------|
| x (mm)        | 25         |
| y (mm)        | 13         |
| heading (deg) | 2          |

TABLE I: Localization Accuracy

1) Localization accuracy: First, we placed Scribbles at a set of known positions on the whiteboard, measured by hand with a ruler relative to the board corners, and compared these ground-truth coordinates against the positions reported by Specs. Second, to measure localization accuracy during path following, we attached a marker to the underside of Specs to trace out the path it follows as it tracks a square path. While executing the path following, localization data was saved. Then, the drawn path was photographed and processed into x-y coordinates. The drawn paths and path from localization were scaled and overlayed onto each other. Then, for each point in the drawn path, the closest point on the path from localization data was found. The distances between the closest points were averaged to give a mean localization distance error.

2) Drawing/Erasing Efficacy: We evaluated the system's ability to reproduce paths by providing a set of standard geometric primitives and line-art patterns. Efficacy was assessed based on the visual continuity of the strokes and the structural integrity of the reproduced shapes. We specifically monitored for mechanical artifacts, such as "jitter" during holonomic transitions or line-width inconsistencies caused by the marker's contact angle. By comparing the physical output to the input trajectory, we assessed the rover's ability to maintain constant contact with the board and execute precise start-and-stop commands without bleeding or skipping ink. To gain quantitative insight on path tracking performance, the traced images were compared to the desired images. Similar to how the actual and localized paths were compared for measuring localization efficiency, we compared the actual path followed by Scribbles to the desired path to calculate a mean position error between the desired and actual image path.

3) System integration: We assessed qualitative robustness of the integrated system, focusing on end-to-end latency, reliability of the wireless link, and behavior under fault conditions such as temporary occlusion of the rover or loss of the Wi-Fi link.

#### B. Localization Results

The homography-based localization using April tags performed reliably whenever the full whiteboard and the rover marker were visible to the PiCam. Mean per-axis error on the board plane was on the order of a few millimeters, which is substantially smaller than the width of the eraser and therefore sufficient for coverage planning I. When drawing actual paths, localization performed better, achieving a mean 5.2 mm of error in position. A comparison of the path calculated from April tag localization with the actual path of Scribbles is provided in Figure 4.

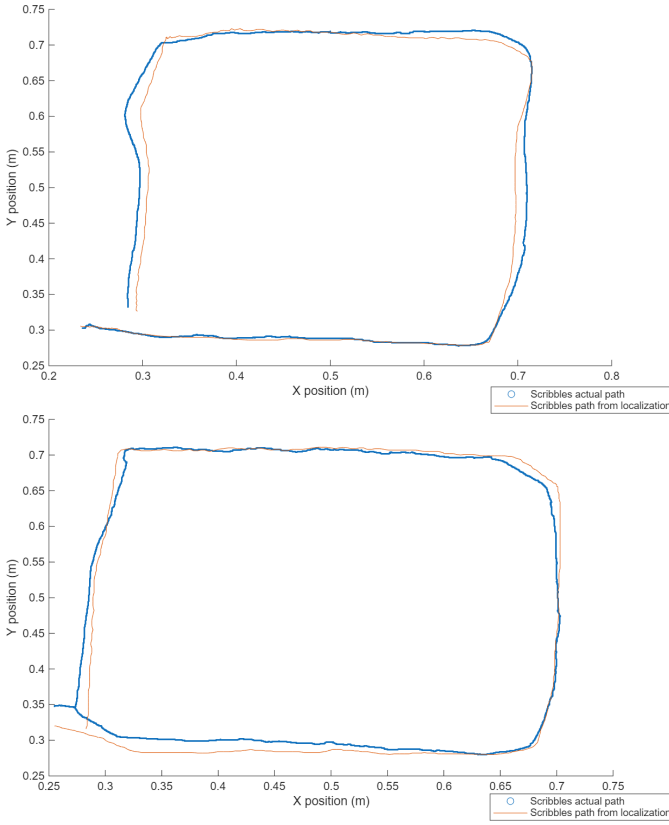


Fig. 4: The localization estimate of Scribbles' position (orange) and actual path followed by Scribbles (blue) for two rectangular paths.

### C. Path following performance

We tested and tuned Scribbles' path following on a square course. Scribbles was able to follow the path with an accuracy of 0.02m, as displayed in figure 5. This path was calculated from the path drawn out by the marker beneath Scribbles. The heading tracking was slightly worse: the heading error reached up to 15 degrees. This affected path following performance as Scribbles would veer off course. One issue that caused the heading offset is that the eraser and marker added friction that required extra force on one side of the robot. As Scribbles moved, the one-sided force from the marker and eraser would shift it round, adding the heading error. Another factor that worsened path following performance was when Scribbles would overshoot waypoints, then be off course before returning to the straight line path between consecutive waypoints. This was mitigated by slowing down when near the desired waypoint. Also, the next waypoint was sometimes sent before Scribbles reached the current waypoint, causing Scribbles to cut corners, leading to the curved corners seen in 5. When we lowered the threshold distance when to send the next waypoint to Scribbles, sometimes Scribbles would be stuck before Specs' localization estimate adjusted to send the next waypoint.

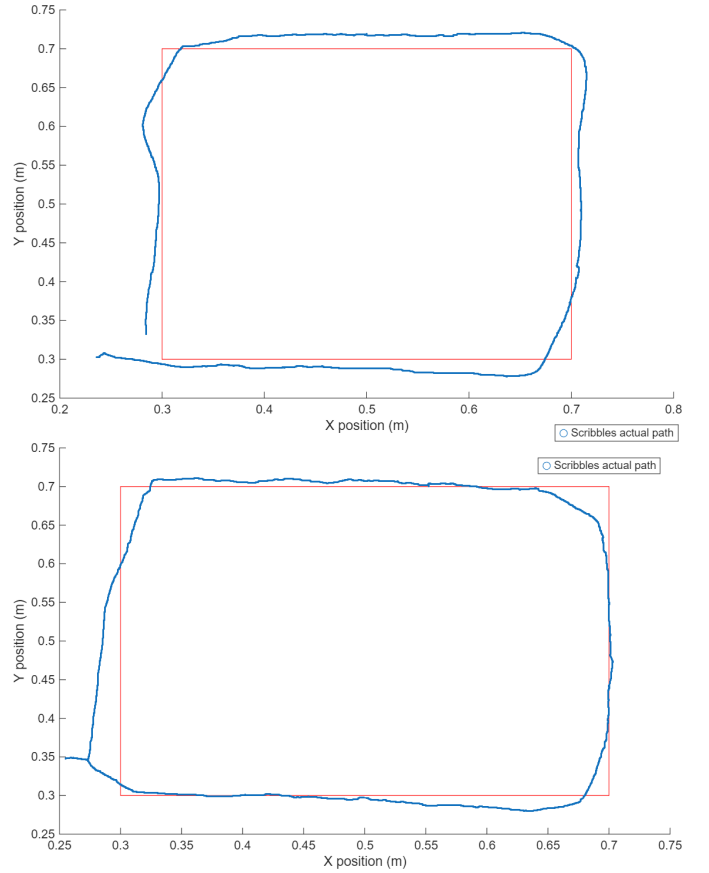


Fig. 5: The path followed by Scribbles (blue) over a desired rectangle (red).

### D. Drawing Efficiency

Scribbles successfully reproduced the input trajectories with decent visual fidelity. The strokes were continuous and dark, confirming that the weight of the rover provided sufficient downward force for the marker without requiring active spring-loading. We observed minor "rounding" at 90° corners, a result of the control loop's deceleration profile, but the overall geometry remained true to the digital source. No significant "marker drift" was observed during long-duration drawing tasks, indicating that the closed-loop localization effectively corrected for any wheel slip.

### E. System Integration

End-to-end latency between a change in rover pose and the corresponding command update was dominated by the vision processing loop on the Raspberry Pi, which was fast enough for the low speeds at which the rover operates. The Wi-Fi link was reliable within the lab environment. The safety timeout on the ESP32 behaved as intended during intentional tests in which the Wi-Fi link was dropped: the rover stopped rather than continuing to drive on its last command. During tests, when communications fell, the rover did not drive out of the whiteboard area.

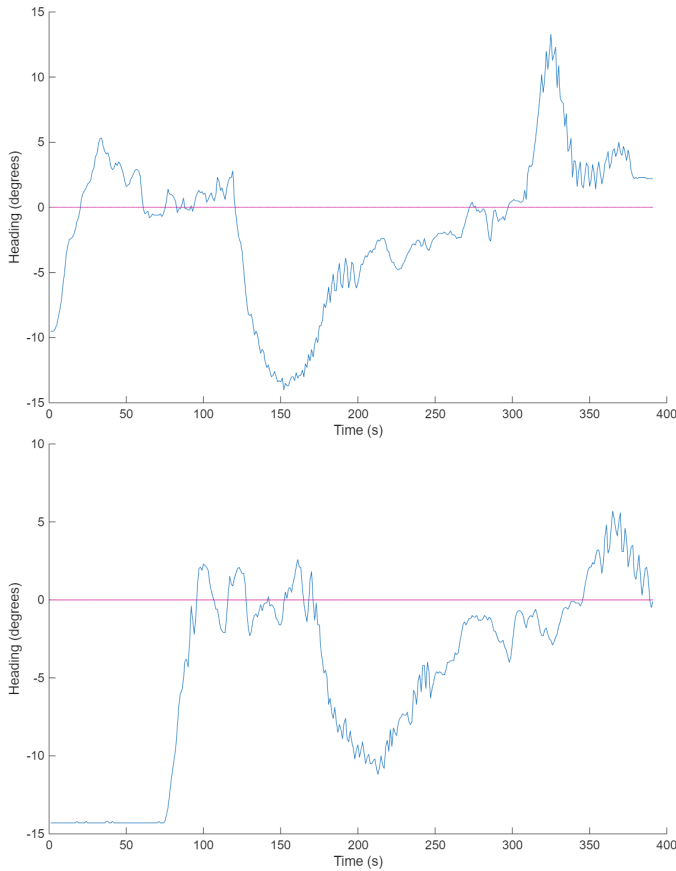


Fig. 6: The heading of Scribbles over a desired rectangle path.

## VI. DISCUSSION AND FUTURE WORK

The most significant deviation from our original proposal was the move from a vertical to a horizontal whiteboard. Our initial design used an array of neodymium magnets beneath the chassis to hold the rover against a ferrous-backed whiteboard. In practice, the combination of magnetic adhesion and our three-wheel omni drivetrain proved unworkable: omni wheels trade lateral friction for holonomic freedom, and even after several iterations, including adding magnets, adjusting the distance between magnet and white board, and re-balancing the chassis, the rover slid downward under its own weight rather than staying in place. Rather than replace the drivetrain (which would have compromised the holonomic motion that simplifies planning) we chose to operate on a horizontal board, which preserves all perception, planning, and communication subsystems.

This experience suggests several directions for future work. First, for a vertical-surface version of the platform, replacing omni wheels with a tracked drivetrain carrying embedded magnets — similar to the design in [3] — would likely give the high-friction, high-normal-force contact needed to climb while still allowing the external vision module to handle localization. Second, the platform is naturally extensible to erasing as well as drawing: replacing the marker holder with an eraser and substituting rasterized user-supplied paths for the erasing tra-



Fig. 7: Input image (top) vs. Image drawn by robot (bottom)

jectory which would turn Scribbles into an autonomous eraser. Another future direction is to adjust control performance for the marker and eraser payloads, as the parameters for control of Scribbles were calculated and tuned without payloads.

## VII. DEMO VIDEO AND CODE

Demo video: [Video](#)

Code repository: [Code](#)

## VIII. STATEMENT OF CONTRIBUTION

Cindy: Designed and assembled Specs platform. Coded all ROS nodes on Specs including ink detection, waypoint generation, communication with ESP32, april tag detection and localization. Also coded multiple test scripts to output numerical results for the report. System integration, report writing, and poster.

Teja: Designed and built several versions of the Scribbles platform with help from Abhi and Art, worked on the report, and did general work helping debugging with the team.

Art: assisted Teja with CAD of Scribbles and construction. Wired up motors and encoders on Scribbles. Wrote code for path following and motor speed control on Scribbles. Tested and tuned control performance. Wrote sections on control and path planning in report. Helped with debugging software during entire system integration.

Abhi: Worked with Teja on mechanical design, as well as general debugging & manufacturing help. Produced project demonstration video and final presentation slides.

## IX. REFERENCES

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